

Himanshu Prajapati

himanshu.codespace157@gmail.com | +91 63060 28308 | [LinkedIn](#) | [GitHub](#) | [Codeforces](#) | Gurugram, India

SUMMARY

Early-career **AI infrastructure engineer** focused on the **LLM inference stack**: open-source contributor to **llm-d**, **vLLM (GuideLLM)**, and **SkyPilot** across inference simulation, request routing, benchmark scheduling, and GPU provisioning; building **eightserve**, a from-scratch serving engine with **continuous batching** and a **paged KV cache**; deployed **Llama 3.1 via vLLM + KServe** on **Kubernetes**. **Competitive programmer**, **500+ problems solved**.

EXPERIENCE

Xponentium

AI Software Intern

Onsite

Jan 2026 - Jul 2026

- **End-to-End Data Migration**: Single-handedly migrated a production ATS from Airtable to PostgreSQL/Supabase, designing a normalized 39-table schema (450+ columns, 38 FKs, ~70K rows) with 55+ versioned SQL migrations.
- **Incremental Sync Service**: Built a Node.js ETL service using modification-time cursors and live schema introspection, cutting DB round-trips **88%**, verified non-destructive by idempotency tests on **14K rows**.
- **Concurrency & Integrity**: Wrote PL/pgSQL RPCs using transaction-scoped advisory locks to eliminate double-booking, and diagnosed an infinite sync loop that cut runs from **20+ minutes to seconds**.
- **Multilingual News Topic Clustering**: Clustered **8,900+ Hinglish/Devanagari** news titles using **LaBSE** with **UMAP + HDBSCAN**, benchmarking **6 architectures** to cut noise from **27.7% to 0%**.

Bithive Technologies

Software Development Intern

Remote

Apr 2025 - Nov 2025

- Built the core frontend in Next.js 14/TypeScript (50+ reusable components), integrated payment gateways and REST APIs, and shipped the Resume Builder and Job Management Dashboard modules.

FunctionUp (YC-23)

Software Engineering Intern

Remote

Jan 2024 - Mar 2024

- Built a serverless WhatsApp notification scheduler on AWS Lambda, EventBridge, and S3 (Python/Flask + Appsmith admin), cutting manual coordination time by 40% at 99.9% reliability.

OPEN SOURCE

llm-d | Distributed LLM Inference (Kubernetes) | Merged Contributions

2026

- Contributed **4 merged Go changes**: in **llm-d-inference-sim**, hardened the tool-call argument generator against under-specified JSON Schemas and gated simulator startup on its native tokenizer sidecar; in **llm-d-router**, fixed the EPP dropping prompt and cached-token usage from streamed Anthropic responses; also authored a namespace `cleanup` subcommand for **llm-d-benchmark**.

vLLM (GuideLLM) | LLM Benchmark Scheduler | Merged Contributions

2026

- Contributed **2 merged Python changes** to the multi-process benchmark scheduler: demand-gated request pulling so saturated workers no longer over-fetch and starve idle ones under burst load, and correct worker-0 attribution in scheduler metadata.

SkyPilot | Multi-Cloud GPU Orchestration | Contributions

2026

- Authored **5 Python changes** across Kubernetes GPU scheduling (directional accelerator-variant matching), **Nebius InfiniBand on Blackwell (B200/B300)**, managed-jobs terminal-state preservation on cleanup failure, idempotent Slurm SSH hooks, and multi-cloud staging buckets.

PROJECTS

eightserve: LLM Serving Engine from Scratch | Python, MLX, FastAPI, asyncio | [GitHub](#)

Ongoing

- **Continuous Batching + Paged KV Cache**: Building an **OpenAI-compatible** engine for **Qwen2.5-1.5B (4-bit)** on an 8 GB M1: a scheduler that admits, batch-decodes, preempts, and evicts every step, and a hand-written **block-based KV allocator** (16-token blocks, block tables, **recounted prefix sharing**) behind a swappable backend vs. a padded-batch baseline.
- **Serving & Measurement**: One engine thread owns all MLX calls, streaming tokens to per-request **asyncio** queues over **SSE**; a frozen harness records **TTFT**, per-user/aggregate **tok/s**, and peak unified memory for 1 to 8 users (~3.3K lines).

Agentic RAG Documentation Assistant | Kubeflow Pipelines, Milvus, KServe, vLLM | [GitHub](#)

- **Four-Layer RAG Architecture**: Built a **Kubeflow** docs assistant (Router → KFP ingestion → **Milvus** → serving), embedding **2,184+ chunks** with citations to hit **91% Precision@3** on a **150-question** benchmark.
- **Inference & Serving**: Deployed **Llama 3.1-8B** via **KServe + vLLM** with tool calling over **WebSocket** and **FastAPI/SSE**, plus an **AST-based** code parser.

Drug Discovery Multi-Agent System | LangGraph, MCP, GPT-4o, Gemini 2.0 | [GitHub](#)

- Built a **LangGraph** orchestrator coordinating Planning, Synthesis, and Validation agents over **MCP** connectors to DepMap, Pharos, and Open Targets, with a **5-layer memory system** for context persistence.

EDUCATION

Polaris School of Technology (Starex University)

B.Tech. in Computer Science and Engineering (8.82 CGPA)

Gurugram, Haryana

2023 - 2027

SKILLS

Languages: Python, Go, C++ (STL), TypeScript, SQL, Bash

LLM Inference & Serving: vLLM, llm-d, KServe, GuideLLM, MLX, Continuous Batching, Paged KV Cache, Prefix Caching, SSE/WebSocket Streaming

Infrastructure & Cloud: Kubernetes, Helm, Kustomize, Docker, SkyPilot, Slurm, Kubeflow Pipelines, AWS, GCP, Linux, CI/CD, Git

AI & Agents: LangGraph, LangChain, MCP, RAG, Multi-Agent Systems, Tool Calling, Milvus, LangSmith Tracing

ML & NLP: PyTorch, Hugging Face, Sentence-Transformers, HDBSCAN/UMAP, Scikit-learn, Pandas

Backend & Concurrency: FastAPI, asyncio, Node.js, PostgreSQL, Supabase, ETL Pipelines, REST APIs, Advisory Locks, Multiprocessing

ACHIEVEMENTS & WRITING

Competitive Programming: Solved **500+ problems** across Codeforces, LeetCode, CSES, and SPOJ; **1000 rating** on Codeforces.

Technical Writing: [KV Cache: A Necessary Evil](#), [KV Cache Part 2: One GPU to a Cluster](#), [A Gigabyte to Write One Word](#),

[AST-Based Chunking for Code RAG](#), [Training vs. Serving](#), [Medium](#)